

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	Proteins
Species Used	Monkey, CV-1, kidney; Mouse, 3T3, embryo; Mouse, p3x63AG8, myeloma; Human: HeLa, epithelial carcinoma		

Before the Pulse

Cell Growth Medium	DMEM (GIBCO/BRL, Sigma)	Growth Phase at Harvest	50 to 60% confluent cells
		Pre-pulse Incubation	5 minutes
Wash Solution	(See notes)		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	0 °C		
Electroporation Medium*	(See notes)	Cuvette Gap	0.4 cm
Cell Density	5 x 10 (6) cells / ml	Voltage	0.5 to 0.75 kV
Volume of Cells	0.2 µl	Field Strength	1.25 to 1.875 kV/cm
DNA Concentration	Not given		
DNA Resuspension Buffer	Not given	Capacitor	25 µF
Volume of DNA	Not given	Resistor	(Pulse Controller) Ω none
		Time Constant	2 to 4 msec

After the Pulse

Outgrowth Medium Pore-Resealing Buffer (see notes) for 10 min. ,37° C; follow with DMEM plus 10% Fetal Calf Serum.

Outgrowth Temperature	37° C
Length of Incubation	Not given
Selection Method or Assay Used	Not given
Electroporation Efficiency	Not given
Per Cent Survival	Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Ref: M.R. Mitchell et. al. (1988) *Experientia* **44**: 199-203.
M.R. Michel et. al. (1990) *J. Virol.* **64**: 5123-5131.
Elgizoli, M. et. al. (1989) *J. Virol.* **63**: 2921-2928.
Electroporation Buffer: 20 mM PIPES, pH 7; 128 mM K-glutamate, 5mM ATP, 5 mM GTP, 10 µm Ca-Acetate, 2 mM Mg-Acetate & amino acid concentration corresponding to that of DMEM media.
Pore-Resealing Buffer: 20 mM PIPES, pH 7.0, 128 mM K-glutamate, 10 µm Ca-Acetate, 2 mM Mg-Acetate

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Survey Number

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